

Unit 7 Pre-Test

Scatterplots & Two-Way Tables

Standards assessed: NC.8.SP.1–SP.4

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.SP.1 Describe patterns of association in bivariate data.

• *I can describe associations in scatterplots and name their key features.*

Match each vocabulary term to its definition, then answer each question.

– v scatterplot	A. As one variable increases, the other tends to increase.
– v positive association	B. A data point that lies far away from the rest of the data.
– v negative association	C. A graph that plots paired data as points.
– v outlier	D. As one variable increases, the other tends to decrease.
– v line of best fit	E. A straight line drawn to model the trend in a scatterplot.
– v cluster	F. A group of data points bunched closely together.

1. Association

As study time increases, test scores increase. Positive or negative association?

Answer: _____

2. Association

As a car's age increases, its value decreases. Positive or negative?

Answer: _____

3. Vocabulary

A point lying far from the pattern is called a(n) ____.

Answer: _____

5. Association

If x and y show no pattern at all, there is ____ association.

Answer: _____

4. Vocabulary

A graph of paired data points is called a ____.

Answer: _____

6. Vocabulary

A line drawn to model the trend of a scatterplot is the line of ____.

Answer: _____

SCORE 4.0 Target Content

NC.8.SP.3 Use a line of best fit to predict, and read two-way tables.

- I can use a line of best fit to make predictions and read values from a two-way table.

Use the line of best fit and the two-way table to answer each question.

Two-way table — use for the questions below

	Likes Pizza	Does Not	Total
Boys	12	8	20
Girls	15	5	20
Total	27	13	40

7. Predict

Line of best fit $y = 2x + 5$. Predict y when $x = 10$.

y = _____

8. Predict

Line of best fit $y = 2x + 5$. Predict y when $x = 0$.

y = _____

9. Interpret

In $y = 2x + 5$, by how much does y increase for each increase of 1 in x ?

=

11. Two-way table

From the table, how many girls were surveyed?

=

10. Two-way table

From the table, how many students like pizza in all?

=

12. Two-way table

From the table, how many students were surveyed in total?

=

SCORE **5.0** Advanced Content

NC.8.SP.4 Interpret relative frequencies and the meaning of slope in context.

- I can find relative frequencies from a two-way table and interpret slope in a real situation.

Use the two-way table above and each best-fit line to answer.

13. Relative freq.

From the table, what percent of boys like pizza?

=

15. Predict

Best-fit line for plant height: $y = 3x + 2$. Predict the height at $x = 6$ weeks.

$y =$

14. Relative freq.

From the table, what percent of girls like pizza?

=

16. Vocabulary

A scatterplot rises left to right, but one point sits far below the rest. That point is a(n) ____.

Answer:

.....

17. *Interpret*

Value vs. age: $y = -4x + 50$. By how much does the value drop each year?

=

.....

.....

18. *Predict*

Line of best fit $y = 1.5x + 10$. Predict y when $x = 8$.

$y =$

.....